Counterexample Packet

A smooth perturbation can cancel an irregular path

Run `fa_banach_001`, lane 1

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Status: candidate full counterexample, likely valid
Target: the smooth-perturbation conjecture in arXiv:1205.1735

1 Source and exact conjecture

For a continuous path $w : [0,T] \to \mathbb{R}^d$, Catellier and Gubinelli define

$$\Phi^w_{s,t}(\xi) = \int_s^t e^{i\langle \xi, w_r\rangle}\, dr, \qquad \|\Phi^w\|_{\mathcal{W}^{\rho,\gamma}_T} = \sup_{\xi\in\mathbb{R}^d} \sup_{0\le s<t\le T} (1+|\xi|)^\rho \frac{|\Phi^w_{s,t}(\xi)|}{|t-s|^\gamma}. \tag{1}$$

The path is (ρ,γ)-irregular when this norm is finite. On PDF page 6, the source conjectures that adding any smooth φ preserves the same pair (ρ,γ) [1].

> An open problem is, for example, understanding what happens if we replace w with a regularised version w^ε or with a perturbed version. In this respect we conjecture that if w is (ρ,γ)-irregular then for any smooth function $\varphi \in C([0,1];\mathbb{R}^d)$ the perturbed path $w^\varphi = w + \varphi$ is still (ρ,γ)-irregular. In relation to this last problem we have obtained the following general result:

Source evidence. The displayed crop is the complete printed conjecture on source PDF page 6.

2 Counterexample

Theorem 1 (Linear cancellation)**.** *For every $0 < \rho < \frac{1}{2}$, set $\gamma = 1-\rho$. On $[0,1]$ in dimension one, let*

$$w(r) = r, \qquad \varphi(r) = -r. \tag{2}$$

Then w is (ρ,γ)-irregular and φ is smooth, but $w+\varphi$ is not (ρ,γ)-irregular. In fact the constant path $w+\varphi$ is not (ρ',γ')-irregular for any $\rho' > 0$ and any $\gamma' > 0$.

Proof. Fix $0 \le s < t \le 1$ and write $h = t-s$. For $\xi \ne 0$,

$$\Phi^w_{s,t}(\xi) = \frac{e^{i\xi t} - e^{i\xi s}}{i\xi}, \qquad |\Phi^w_{s,t}(\xi)| \le \min\left\{h, \frac{2}{|\xi|}\right\}. \tag{3}$$

If $|\xi| \le 1$, then, using $1-\gamma = \rho$,

$$(1+|\xi|)^\rho \frac{|\Phi^w_{s,t}(\xi)|}{h^\gamma} \le 2^\rho h^{1-\gamma} = 2^\rho h^\rho \le 2^\rho. \tag{4}$$

This also covers $\xi = 0$. If $|\xi| > 1$, the elementary interpolation bound $\min\{A, B\} \le A^{1-\rho}B^{\rho}$ applied to (3) gives

$$|\Phi^w_{s,t}(\xi)| \le 2^{\rho}h^{1-\rho}|\xi|^{-\rho} = 2^{\rho}h^{\gamma}|\xi|^{-\rho}.$$

Consequently

$$(1+|\xi|)^{\rho}\frac{|\Phi^w_{s,t}(\xi)|}{h^{\gamma}} \le 2^{\rho}\left(\frac{1+|\xi|}{|\xi|}\right)^{\rho} \le 4^{\rho}. \tag{5}$$

Equations (4)–(5) prove $\|\Phi^w\|_{\mathcal{W}_1^{\rho,\gamma}} \le 4^{\rho} < \infty$. Since $\gamma = 1-\rho > \frac{1}{2}$, this also makes w ρ-irregular in the source's terminology.

On the other hand, $z := w + \varphi \equiv 0$, so

$$\Phi^z_{s,t}(\xi) = t - s \tag{6}$$

for every ξ. Taking $s = 0, t = 1$ in (1) yields

$$\sup_{\xi\in\mathbb{R}}(1+|\xi|)^{\rho'}|\Phi^z_{0,1}(\xi)| = \sup_{\xi\in\mathbb{R}}(1+|\xi|)^{\rho'} = \infty$$

for every $\rho' > 0$. Thus z has no positive irregularity exponent, proving the claim and refuting the printed conjecture. □

3 Why the example is decisive

The conjecture is universally quantified over (ρ, γ)-irregular paths and over smooth perturbations. One admissible pair and one smooth perturbation therefore suffice. The construction is not exploiting an endpoint: ρ may be any number strictly between 0 and 1/2, while $\gamma = 1-\rho$ is strictly greater than 1/2.

The mechanism is complete cancellation. A nonconstant smooth curve already has a small positive irregularity exponent because its oscillatory integral decays like $|\xi|^{-1}$ after the time interval is resolved. The conjecture allows a smooth perturbation depending on that curve, so the perturbation can cancel it exactly.

4 Verification and scope report

- **Definition check.** The proof uses the source's inhomogeneous factor $(1+|\xi|)^{\rho}$; (4) handles the low-frequency regime explicitly.
- **Time exponent.** The interpolation in (3) produces exactly $h^{1-\rho}$, so choosing $\gamma = 1-\rho$ gives the required uniform bound.
- **Smoothness.** Both w and φ in (2) are C^{∞}. The conjecture places no independence or roughness restriction on them.
- **Failure strength.** The perturbed path fails every positive spatial irregularity exponent, not merely the original ρ.
- **Scope limitation.** This refutes the literal statement through the range $0 < \rho < 1/2$. It does not decide whether smooth perturbations preserve (ρ, γ) for suitably rough paths or for $\rho > 1/2$.

5 Later literature and human review

The run's registry, solution, attempt, and proof-gap indexes contain no prior record for arXiv:1205.1735 or this counterexample. A bounded primary-literature search found the latest source version and Galeati–Gubinelli's later study [2]. Section 5.4 of that paper calls the smooth-additive assertion a conjecture left open in the source, assumes $\rho > 1/2$ for its additive analysis, and proves a positive stability result only after strengthening the hypothesis to *strong* ρ-irregularity. It does not state the linear cancellation family above. Novelty confidence is therefore moderate, not absolute.

Before any external claim, a specialist should check whether the source authors intended an unstated restriction such as $\rho > 1/2$, strong irregularity, or a perturbation prescribed independently of w. None appears in the printed conjecture reproduced above.

References

[1] R. Catellier and M. Gubinelli, *Averaging along irregular curves and regularisation of ODEs*, arXiv:1205.1735v4.

[2] L. Galeati and M. Gubinelli, *Prevalence of ρ-irregularity and related properties*, arXiv:2004.00872v2.